SWIZ SMART PROTOCOL

Technical Specification, Contract Functions & Application Modules

An exhaustive engineering guide to the 20-year locked vault, algorithmic dynamic pricing formula,

EXECUTIVE SYSTEM SUMMARY & ARCHITECTURAL HIGHLIGHTS

Protocol Name: Swiz Smart Protocol (Swiz Token / SZ). Production Testnet: Polygon Amoy (Chain ID: 80002).
Core Token Economy: Swiz Token (SZ, ERC-20, 6 decimals, fixed 1B supply in contract custody), Ecosystem Token (ET, ERC-20, 6 decimals, minted on Year 2-20 buybacks), Wrapped USDT (USDT.c, 6 decimals, 1:1 cross-chain deposit receipt token).
20-Year Linear Staking Schedule: Every purchase or offline sale locks tokens into custody and unlocks exactly 5.00% annually (365-day intervals) across 20 distinct tranches.
Compounded Algorithmic Pricing: Algorithmic protocol reference price climbs +2.33% compounded per 1,000 completed packages and per $100,000 USD cumulative new staking volume. Computed purely on-chain without oracle risk.
Fully Automated FIFO Buyback Engine: Unlocked tranches queue in an on-chain priority min-heap. When new buyers purchase SZ, the contract matches eligible stakers automatically, with zero claim transactions required from users.
Rule-Enforced Multi-Asset Payout: Year 1 tranches pay 100% USDT cash directly to original stakers. Years 2-20 tranches issue 100% ET tokens to stakers, routing the incoming USDT payment directly to Corporate Treasury.
Atomic Bulk Offline Ingestion: Native contract function recordOfflineSalesBatch executes up to 100 offline purchaser allocations in a single MetaMask signature, saving 90%+ in gas fees.

SYSTEM METRICS & ACTIVE PROTOCOL REGISTRY

Parameter / Contract	Address / Value	Architecture Role
SZToken Contract	0x2cfac91e27B9f9c855D4d8f248Da28Ef54c54d47	Core protocol vault, pricing engine, 20-yr tranches, FIFO mat
EcosystemToken (ET)	0x32105EF98AD3473036eF614C835a81475E27d06E	Secondary reward asset. Minter permanently bound strictly to
USDT.c Wrapper	0xAfECBA18585f6C9eb120fD89a56B53243856a4fB	Cross-chain 1:1 wrapped receipt asset on Polygon Amoy
Source Deposit Adapter	0x098DeA8e03381E8187F472308004530D52E5A0df	Inbound bridge adapter receiving USDT and emitting verified
Mock USDT (Amoy)	0x9effc8c62f9077EDB785E6482328A7BfD7eAbe9f	6-decimal purchase collateral token used for purchases & Ye
Company Treasury (AW)	0x1Eb5FDFa920aE1E8341288e4F69c698698FC5926	Receives buyer USDT payments from Year 2 onward and ex
Protocol Admin Wallet	0x743B5509Cc84415a9E1E20951E2fD94d05E423e9	Authorized DEFAULT_ADMIN_ROLE & ADMIN_ROLE on al
Total Fixed Supply	1,000,000,000 SZ (1 Billion)	Fixed supply. Zero administrative minting, burn disabled

SECTION 1: HIGH-LEVEL ARCHITECTURE & ECOSYSTEM TOPOLOGY

The Swiz Smart protocol is a non-custodial, decentralized progressive staking and automated liquidity matching ecosystem. It links a multi-chain deposit gateway, an EVM execution engine on Polygon Amoy, and an enterprise React 19 dApp with real-time on-chain synchronization.

1.1 THREE-TIER ARCHITECTURE OVERVIEW

Architecture Layer	Primary Technology Stack	Responsibilities & Core Capabilities
Frontend Presentation	React 19, TypeScript, Vite, TailwindCSS, Ethers.js v6, Recharts, lucide-react	User portfolio tracking, 20-yr unlock schedule visualizer, bridge UI, CSV bulk offline
Middleware & Indexer	Node.js, Express, Ethers.js, SQLite / Redis Store	Bridge relayer coordination, cross-chain deposit event listener, secondary wallet reg
On-Chain Smart Contracts	Solidity 0.8.28, OpenZeppelin v5, Hardhat, Polygon Amoy	Fixed-supply custody vault, algorithmic dynamic pricing, priority min-heap FIFO buy

1.2 MULTI-TOKEN TOPOLOGY & ASSET ROLES

Token Asset	Standard / Decimals	Minting Authority	Functional Role in Protocol
Swiz Token (SZ)	ERC-20 (6 Decimals)	Pre-minted (Fixed 1B Supply)	Locked protocol asset. Held in contract custody; unlocks 5% a
Ecosystem Token (ET)	ERC-20 (6 Decimals)	Exclusively Bound to SZToken	Reward token issued to stakers for Year 2 to 20 buybacks. Ze
Wrapped USDT (USDT.c)	ERC-20 (6 Decimals)	USDTcWrapper Relayer Role	1:1 receipt token minted on Polygon Amoy against verified sou
Collateral USDT (Amoy)	ERC-20 (6 Decimals)	OpenZeppelin ERC-20 Mock / Native	Official collateral used for purchase payments, Treasury reserv

1.3 WALLET MODEL & PRIVILEGE HIERARCHY

- **User Wallet (UW): Standard community wallet. Purchases packages, receives automated Year 1 USDT payouts and Year 2-20 ET** tokens directly without claims.
- **Corporate Treasury / Admin Wallet (AW): Company-controlled EVM address receiving Year 2-20 buyer USDT payments, price cap** spreads, and fraction-cent rounding dust.
- **Secondary / Backup Wallet: User-designated recovery address configured in DApp. If UW is compromised, protocol admin can** invoke replaceWallet() to safely rescue all staking packages.

SECTION 2: FRONTEND APPLICATION MODULES & USER FLOWS

The frontend is built with high modularity and enterprise UX resilience. All contract reads and writes utilize ethers.js v6 with automatic RPC fallback across Alchemy, Polygon Bor, and public endpoints.

2.1 USER DASHBOARD & PORTFOLIO TRACKER (DashboardPage.tsx)

- **Live Asset Overview: Queries user balances in parallel: locked SZ in protocol custody, claimable/unlocked SZ, free ET** token balance, and USDT liquidity.
- **20-Year Tranche Schedule: Fetches getUnlockSchedule(pkgId) for each active package. Visualizes 20 distinct annual tranches** with exact countdown timestamps, unlock percentages (5% each), and status badges (Locked, Available for Buyback, or Bought Back).
- **Secondary Wallet Linking: Displays connected secondary backup wallet status. Alerts users if an active secondary recovery** wallet has been bound to their account.

2.2 TOKEN PURCHASE & STAKING MODULE (BuyTokensSection.tsx)

- **Pre-Flight Validation: Enforces protocol bounds (minPurchase: 1 USDT, maxPurchase: 100,000 USDT). Verifies user USDT** balance and allowance.
- **Dynamic Price Execution: Calls getCurrentPrice() to calculate exact SZ allocation (qty = value * 1e6 / price).** Automatically sets deadline (now + 600s) and slippage protection minSZ.
- **Optimized Gas Profiling: Employs an intelligent gas ceiling (750,000 gas) preventing MetaMask from demanding excessive POL** balances (~0.015 POL actual fee).

2.3 CROSS-CHAIN USDT BRIDGE PORTAL (DepositPage.tsx)

- **Multi-Chain Inbound Gateway: Allows stakers on Ethereum Sepolia (Chain 11155111) and BSC Testnet (Chain 97) to deposit** USDT into SourceDepositAdapter.
- **Direct Treasury Custody: Adapter immediately sweeps deposited USDT to Corporate Treasury wallet on the native chain while** emitting a Deposited event with a monotonic nonce.
- **Atomic Minting On Destination: Bridge coordinator validates confirmations and triggers USDTcWrapper.mintDeposit() on** Polygon Amoy, delivering 1:1 USDT.c to the depositor.

2.4 BULK OFFLINE SALES UPLOADER (ProtocolOperations.tsx)

- **Single-Signature Batch Execution: Admin can upload a CSV containing up to 100 offline sales and execute** recordOfflineSalesBatch in 1 transaction, saving 90%+ gas.
- **Real-Time Duplicate Reference Check: Parses CSV and queries contract recordedReferences(hash(ref)) for every row. Invalid** or previously used references trigger instant red UI warnings before broadcasting.
- **Checksum Sanitization: Auto-normalizes and checksums buyer wallet addresses via ethers.getAddress to prevent** case-sensitive reverts.

SECTION 3: CORE CONTRACT DEEP DIVE - SZTOKEN.SOL

SZToken.sol (0x2cfac...4d47) is the central authority of the protocol. It combines ERC-20 fixed token custody, OpenZeppelin v5 AccessControl (RBAC), ReentrancyGuard, Pausable, and the dynamic FIFO buyback matching engine.

3.1 STATE STRUCTURES & DATA MODELS

SZToken.sol - Storage Structs

```
struct Sale { address buyer; uint256 amount; uint256 value; uint256 price; uint256 purchasedAt; uint256 startedAt; bytes32 refe
struct Tranche { uint256 unlockTimestamp; uint256 quantity; uint8 status; uint256 boughtBack; } // 0=Locked, 1=Available, 2=Bou
struct Package { uint256 id; address owner; uint256 amount; uint256 originalPrice; uint256 startTimestamp; bool isOffline; uint
struct Position { uint256 unlockAt; uint256 packageId; uint256 tranche; } // Priority Queue item
```

3.2 ALGORITHMIC DYNAMIC PRICING FORMULATION (getCurrentPrice)

The algorithmic protocol reference price starts at exactly 1.00 USD (1,000,000 units with 6 decimals) and climbs by a compounded 2.33% (multiplied by 10233 / 10000) whenever protocol milestones are crossed:

- **Package Counter Step: Increments price exponent for every 1,000 completed packages (both online purchases and verified** offline sales).
- **Staked Volume Counter Step: Increments price exponent for every $100,000 USD (100,000 * 10^6) in cumulative staked** purchase volume.
- **Compounded Exponent: Formula: k = (totalPackages / 1000) + (totalStakedUSD / 100,000e6). Price = initialPrice *** (1.0233)^k.
- **Deterministic View: Calculated deterministically on-chain without price oracle dependencies, eliminating flash-loan** manipulation risk.

3.3 20-YEAR UNLOCK TRANCHE MATHEMATICS

- **Linear Annual Allocation: Tranches 0 through 18 unlock exactly amount / 20 (5.00% of the total package) every 365 days.**
- **Final Tranche Dust Absorption: Tranche 19 (Year 20) unlocks: amount - (amount / 20) * 19. This guarantees 100.000000% of** the tokens are unlocked without rounding losses.
- **Unlock Eligibility: Tranche t becomes eligible for buyback at: unlockAt = package.startTimestamp + (t + 1) * 365 days.**

SECTION 4: CORE SOLIDITY FUNCTIONS SPECIFICATION

The table and function breakdowns below document the exact signature, permissions, gas characteristics, and side effects of all key methods in SZToken.sol.

Solidity Function Signature	Visibility / Role	Gas Range	Key On-Chain Action
purchase(uint256,uint256,uint256)	External / Public	250k - 8.8M	Pulls USDT, matches FIFO buyback queue, updates price, creates new 2
recordOfflineSale(Sale)	External / ADMIN	~180k - 250k	Validates offline purchase, deducts treasury inventory, creates package.
recordOfflineSalesBatch(Sale[])	External / ADMIN	~400k - 8.5M	Processes up to 100 offline sales atomically in 1 single transaction.
getCurrentPrice()	External / View	~5,000	Returns authoritative compounded algorithmic reference price with 6 dec
getUnlockSchedule(uint256)	External / View	~25,000	Returns full 20-tranche array with unlock dates, quantities, and status.
getUserPackageIds(address)	External / View	~15,000	Returns array of package IDs owned by specified wallet.
replaceWallet(address,address)	External / ADMIN	~120k	Emergency migration: transfers packages to pre-registered secondary wa
setWalletPaused(address,bool)	External / ADMIN	~45k	Freezes/unfreezes a wallet from purchasing or receiving buyback payout
setLimits(uint256,uint256,uint256)	External / ADMIN	~35k	Updates minimum and maximum purchase bounds and package caps.
pause() / unpause()	External / ADMIN	~30k	Emergency protocol-wide circuit breaker halts transfers and purchases.

4.1 THE FIFO MATCHING CASHIER (_match)

When a buyer executes purchase(), the contract enters the private _match() routine before touching unsold treasury inventory:

- **Priority Min-Heap (heap): Queue items Position(unlockAt, packageId, tranche) are sorted by earliest unlock timestamp.** Oldest unlocked tranches are always matched first.
- **Price Cap Rule: Buyback price = min(currentPrice, 100 * package.originalPrice). Protects protocol from excessive** liabilities.
- **Year 1 Rule (Tranche 0): 100% of payout transferred as USDT directly to original staker’s wallet. Staker receives cash** return.
- **Years 2-20 Rule (Tranches 1-19): 100% of payout is minted as Ecosystem Token (ET) directly to staker’s wallet. 100% of** buyer’s USDT is routed to Company Treasury.
- **Treasury Spread Capture: If currentPrice > buybackPrice (due to the 100x cap) or if rounding dust exists, excess USDT is** safely transferred to Treasury.

4.2 AUTOMATED RESCHEDULING (_scheduleNext)

- **Continuous Queue Progression: When tranche t of a package is fully bought back, the contract automatically pushes tranche** t + 1 into the priority heap, provided t < 19 and the tranche is already unlocked.

SECTION 5: SUPPORTING CONTRACTS & CROSS-CHAIN BRIDGE

5.1 ECOSYSTEMTOKEN.SOL (ET) - SECONDARY REWARD ASSET

EcosystemToken.sol (0x32105...d06E) is a specialized standard ERC-20 token (6 decimals) created exclusively to distribute platform value to Year 2-20 stakers.

- **Permanent Minter Binding: Contains bindMinter(address minter). During deployment, it was permanently locked to SZToken** address. Re-binding is strictly impossible.
- **Zero Arbitrary Minting: No administrator, developer, or deployer wallet possesses minting authority. ET can only be** created by SZToken when an on-chain buyback occurs.
- **No Direct Treasury Claim: ET is transferable across EVM wallets and can be utilized in secondary dApp utilities, but** carries no redemption claim against company reserves.

5.2 USDTCWRAPPER.SOL - ONE-WAY MANAGED WRAPPER

USDTcWrapper.sol (0xAfECB...a4fB) is deployed on Polygon Amoy. It issues USDT.c receipt tokens representing validated inbound cross-chain deposits from source EVM chains.

- **Cross-Chain Replay Protection: Maintains mapping(bytes32 => bool) public mintedDeposits. A deposit hash keccak256(chainId,** adapter, depositor, amount, nonce, txHash) can only be minted once.
- **Source Adapter Whitelist: Function configureSource(uint256 chainId, address adapter) ensures only authorized adapter** contracts on verified chains trigger minting.
- **One-Way Flow: USDT.c is minted 1:1 on Amoy. No reverse burn-and-release mechanism exists, eliminating cross-chain drain** vulnerabilities.

5.3 SOURCEDEPOSITADAPTER.SOL - INBOUND PORTALS

Deployed on source networks: Ethereum Sepolia (Adapter: 0x279bF...284F), BSC Testnet, and Polygon Amoy (Adapter: 0x098De...A0df).

- **Immediate Treasury Sweep: When deposit(amount, recipient) is called, source USDT is immediately transferred directly into** Company Treasury on that chain.
- **Monotonic Nonce Generation: Emits Deposited(nonce++, depositor, recipient, amount, block.timestamp) providing immutable** cryptographic evidence for relayer minting.

SECTION 6: SECURITY, ACCESS CONTROL & ON-CHAIN PROOFS

6.1 ROLE-BASED ACCESS CONTROL (RBAC) MATRIX

Role Identifier	Role Hash (keccak256)	Assigned Address	Granted Capabilities
DEFAULT_ADMIN_ROLE	0x0000...0000 (0x0)	0x743B5509...23e9	Full governance: grant/revoke roles, configure global limits, eme
ADMIN_ROLE	0x75b238...48a2	0x743B5509...23e9	Operational authority: record offline sales, execute replaceWalle
Deployer Key	Temporary Setup Signer	0x9B5AfF...9c8E	Initial deployment & ET minter binding. Relinquished operationa

6.2 VERIFIED LIVE TRANSACTIONS ON POLYGON AMOY (CHAIN 80002)

All protocol mechanics have been deployed and verified live on-chain. Below are official transaction proofs visible on Polygonscan Amoy:

Protocol Operation	Transaction Hash / Link	Key Verified Result
Contract Deployment	Block 51195330	SZToken (0x2cfac...), ET (0x32105...), 1B SZ fixed supply minted.
ET Minter Binding	Executed in Deploy Block	ET.bindMinter permanently locked to SZToken contract address.
Admin Role Grants	0x9d7396cf8379...51844	DEFAULT_ADMIN & ADMIN_ROLE granted to 0x743B5509Cc84415a9E1E20951E
Bulk Offline Upload	0xc1c2cc4e21ee...9c8f10	3 offline sales recorded simultaneously in 1 single transaction (Pkgs #2, #3, #4).
100-Tranche Buyback	0x1a5ae40e77ba...ca7e	1,000 SZ purchase matched 100 tranches: 50 USDT paid (Yr 1) + 950 ET minted (Y
Cross-Chain Bridge	0x9d7410b18a42...c3bf	Source USDT deposited to adapter and swept directly into Company Treasury.

SECURITY AUDIT & RESILIENCE ASSURANCES

1. Fixed Supply Invariant: Exactly 1 Billion SZ tokens exist. No mint() or burn() functions exist on SZToken.sol.
2. Non-Decreasing Price: Algorithm guarantees price can only stay constant or compound upwards. Oracle manipulation is impossible.
3. Rounding Dust Protection: Solidity integer division dust is caught and directed to Treasury, protecting protocol solvency.
4. Zero-Claim Passive UX: Stakers never sign claim transactions. Buyback proceeds land in their wallets automatically.